Code No: 136BC **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester (Special) Examinations, January/February - 2021
DESIGN PATTERNS
(Computer Science and Engineering)

Time: 2 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1. What is a Design Pattern? Write the general template in describing a Design Pattern. [15]

2. Give the step-by-step procedure in designing a Document Editor and explain briefly. [15]

3. What are the key consequences and suitability conditions of Prototype and Singleton design pattern? Explain. [15]

4.a) Write down the motivation of Abstract Factory and Builder patterns.
b) Draw the structure of a Factory pattern and explain the responsibilities and roles of its participants. [6+9]

5. Explain in brief about the intent, structure, responsibilities and key consequences of Adapter and Bridge Design patterns. [15]

6. Explain in brief about Chain of Responsibility and Command design patterns. [15]

7.a) State the uses and related patterns of Flyweight design Pattern.
b) What are the implementation issues of Mediator design pattern? [8+7]

8.a) When can be a Façade design pattern used effectively? Discuss briefly.
b) Discuss the motivation and applicability of Memento design pattern. [7+8]

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used paper jan-feb-2021